A positive-power counterexample for Taylor intersection roots

A solution of MathDB 328557

Research notes in Principia Math In Progress

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1 Task 1: the problem and its resolution

For $f(z) = z^n$ and a positive integer $r < n$, let

$$P_c(z) = \sum_{k=0}^{r} \frac{f^{(k)}(c)}{k!}(z-c)^k$$

be the order-r Taylor polynomial at c. Conjecture 2 in Section 5.2 of Horwitz's paper [1] asserts that, when $0 < a < b$, every nonreal root of $P_b - P_a$ has real part strictly between a and b. The following refutes that assertion for a positive integer power.

Theorem 1. *Take $r = 5$, $n = 24$, $a = 1$, and $b = 2$. Then $P_2 - P_1$ has a conjugate pair of nonreal roots z_+, z_- satisfying*

$$2.00042 < \operatorname{Re} z_\pm < 2.00044, \qquad 0.29417 < |\operatorname{Im} z_\pm| < 0.29419.$$

In particular, their real parts are greater than b.

All source hypotheses hold: $0 < a < b$, $r > 3$, $n \geq r + 1$, and $f^{(r+1)}$ is continuous and strictly positive on $[a, b]$. The source's examples for negative powers concern a different extension. The counterexample here uses the positive power explicitly required by Conjecture 2. It does not refute the weaker Conjecture 1, which asks only for at least one conjugate pair with real part between the nodes.

2 The exact polynomial

Put $y = z - 2$. Directly from the binomial coefficients,

$$P_2(2+y) = \sum_{k=0}^{5} \binom{24}{k} 2^{24-k} y^k,$$

$$P_1(2+y) = \sum_{k=0}^{5} \binom{24}{k} (1+y)^k.$$

Their difference is $7Q(y)$, where

$$Q(y) = 3183470664y^5 + 1591706490y^4 + 606309440y^3 + 165304680y^2 + 28723560y + 2388823.$$

Thus it suffices to isolate a nonreal zero of Q in the right half-plane.

3 An exact rational root certificate

Let

$$c = \frac{43 + 29418i}{100000}, \qquad \rho = \frac{1}{100000}, \qquad d_k = \frac{Q^{(k)}(c)}{k!}.$$

Every d_k has rational real and imaginary parts. For a complex number v write $U(v) = |\operatorname{Re} v| + |\operatorname{Im} v|$ and $D(v) = \max(|\operatorname{Re} v|, |\operatorname{Im} v|)$; then $D(v) \le |v| \le U(v)$. Exact substitution gives the following rational inequalities, with deliberately loose integer bounds:

k	upper bound for $U(d_k)$
0	89
2	937000000
3	4028000000
4	6282000000
5	3184000000

$$D(d_1) > 65000000.$$

These are finite integer-arithmetic claims: the companion verifier evaluates each rational real and imaginary part exactly and checks the displayed inequalities. They can also be checked directly from the displayed Q and c.

Since every upper bound in rows 2 through 5 is less than 10^{10},

$$\sum_{k=2}^{5} |d_k| \rho^k < 10^{10} \frac{\rho^2}{1-\rho} < 2.$$

On the circle $|y - c| = \rho$ it follows that

$$|Q(y) - d_1(y-c)| \le |d_0| + \sum_{k=2}^{5} |d_k| \rho^k < 91 < 650 < |d_1(y-c)|.$$

Rouché's theorem therefore gives exactly one zero of Q in $|y - c| < \rho$, counted with multiplicity, because the comparison polynomial $d_1(y - c)$ has one. The disk lies entirely in

$$0.00042 < \operatorname{Re} y < 0.00044, \qquad 0.29417 < \operatorname{Im} y < 0.29419.$$

The coefficients of Q are real, so its conjugate disk contains the conjugate root. Adding 2 gives the assertions of Theorem 1. This closes the proof; decimal root approximations are not used.

4 Follow-up tasks 2–5

Task 2. Determine all pairs (r, n) for which every nonreal intersection root lies in $a < \operatorname{Re} z < b$ for every $0 < a < b$. In particular, find the least positive exponent producing failure for each fixed Taylor order.

Task 3. For fixed (r, n), classify the ratios $b/a > 1$ at which a root crosses either boundary of the vertical strip. Obtain exact algebraic criteria and locate every transition for the case $(r, n) = (5, 24)$.

Task 4. Prove a precise limit theorem for the roots as $n \to \infty$ with r and b/a fixed. Determine how the zeros of truncated exponential series control the number and size of excursions beyond b.

Task 5. Resolve the weaker assertion that at least one nonreal conjugate pair has real part in (a, b) whenever $f^{(r+1)}$ is continuous and nonvanishing there and $r \ge 4$. Identify sharp hypotheses if this assertion also fails.

References

[1] A. Horwitz, *Means and nonreal intersection points of Taylor polynomials*, Journal of Classical Analysis 6 (2015), 1–28, Conjecture 2 in Section 5.2, p. 26. https://doi.org/10.7153/jca-06-01. Preprint: https://arxiv.org/abs/1408.2573.

[2] MathDB 328557, live record retrieved 11 September 2026. https://mathdb.com/p/328557.